FINAL WORKFLOW

PS-3: How do we predict the location, and how sure can we be?

This is the consolidated workflow after the later refinements: Bayesian/grid inference for visited cases, a spatial-constraint approach for the no-visit case, robust handling of bad evidence, residual learning as a research option, and conformal calibration for the final confidence radius.

The one-sentence idea
We start with where the geocoder and other spatial clues suggest the house could be, update that belief using trustworthy field evidence, produce a location probability distribution, turn its spread into a raw confidence region, and then calibrate that region against held-out ground truth.

What is in scope

- Location inference after the upstream NLP/address-parsing team has produced structured address clues.
- Use of geocoder, locality, landmark, field-visit and agent/trail information.
- Handling zero visits, multiple visits, conflicting visits and suspicious visits.
- Prediction of both a point (x,y) and an uncertainty region/radius.
- Comparison of Bayesian, robust, clustering and residual-learning alternatives.

What is NOT in scope

Raw address tokenization, transliteration, abbreviation expansion or address embeddings. Another team is responsible for producing structured address-derived features.

Important mindset
Bayesian + Gaussian is a candidate framework, not a claim of automatic optimality. The dataset itself contains reasons to improve the likelihood, exploit spatial constraints, and compare alternatives.

1. The problem and what the data tells us

The output for every address is two things: (1) a predicted point (x,y), and (2) a radius with a coverage claim such as “80% confidence”. The coordinates are on a local grid measured in metres, so straight-line distance is directly meaningful.

Fact	What we actually know	Why it matters
F1	100 addresses have verified true coordinates; official split = 66 train / 19 validation / 15 test.	Only 66 verified targets are available for fitting supervised relationships; overfitting is a real risk.
F2	Geocoder median error changes sharply with precision: street ~109 m, locality ~386 m, pincode ~1,376 m; rooftop has only 1 surveyed example.	Precision should control how wide the geocoder prior is; rooftop-specific uncertainty cannot be learned reliably from one case.
F3	2,052 / 2,880 geocodes are locality; only 50 are rooftop.	The geocoder is usually a broad locality clue, not a house-level truth source.
F4	1,477 / 3,117 addresses have at least one visit; among the 100 surveyed, 49 have visits and 51 do not.	The no-visit case is a core case, not an edge case.
F5	Door-reaching outcomes are often close to truth: met_family ~16 m, locked_premises ~20 m, neighbour_says_shifted ~20 m, met_borrower ~34 m median.	Good field visits can be much stronger than a locality-level geocoder.
F6	The same visit outcomes have huge tails: locked_premises P90 ~3.9 km; some met_borrower visits are ~7.9 km away.	A pure Gaussian cannot describe the full error distribution well; outliers/heavy tails matter.
F7	address_not_traceable median distance from truth ~1,603 m.	This is better interpreted as failed search / negative evidence, not a normal “house location” measurement.
F8	Visit error rises with GPS accuracy; median error / gps_accuracy_m is ~2.4.	Reported GPS accuracy is a useful signal but is not the same thing as actual location uncertainty.
F9	172 visits share a photo_hash with another visit; agent FA009 has much worse door-visit error than others.	Some visits may be unreliable, so the model needs a visit-trust mechanism.

Bottom line

The data is saying: geocoder is useful but coarse; good visits are powerful; a few visits are wildly wrong; and half the labelled addresses have no visits. The workflow has to be built around those facts.

2. The full mental model

Think of the model as a sequence of evidence updates rather than as one giant ML model.

STRUCTURED ADDRESS FEATURES (upstream team)
+
COMMERCIAL GEOCODER
+
LANDMARK / LOCALITY INFORMATION
+
FIELD VISITS + GPS TRAILS + AGENT SIGNALS
↓
SPATIAL PRIOR
↓
VISIT TRUST / RELIABILITY
↓
VISIT LIKELIHOODS
↓
BAYESIAN / GRID FUSION
↓
POSTERIOR OVER MAP LOCATIONS
↓
PREDICTED PIN
↓
RAW UNCERTAINTY
↓
CONFIDENCE REGION / RADIUS
↓
CONFORMAL CALIBRATION
↓
FINAL PIN + CALIBRATED RADIUS

Two important branches

Case	Main evidence	Natural method
Visits available	Geocoder prior + trustworthy visits + spatial context	Bayesian/grid fusion with robust visit likelihood
No useful visits	Geocoder + locality + landmarks + address type + nearby spatial clues	Spatial-constraint / convex-hull-style prior, then confidence boundary + calibration

Why keep two branches?
Visited addresses have direct location evidence that can sharply update the prior. No-visit addresses do not. Forcing the same machinery on both cases can waste the non-visit spatial clues.

One conceptual distinction to keep throughout

- Prior: where the house initially seems plausible.
- Likelihood: how compatible a particular visit is with candidate location L.
- Posterior: the updated belief over locations after all evidence is combined.
- Uncertainty: how spread out that posterior belief is.
- Confidence radius: how much area we keep around the predicted point for a chosen coverage level.
- Conformal calibration: checks and adjusts that region using held-out truth.

3. Step 1 - represent the map as candidate locations

For the Bayesian/grid branch, we represent the plausible search area as small cells, e.g. 25 m × 25 m. Each cell is a candidate location L.

```
+-----+-----+-----+-----+
| L1  | L2  | L3  | L4  |
+-----+-----+-----+-----+
| L5  | L6  | L7  | L8  |
+-----+-----+-----+-----+
| L9  | L10 | L11 | L12 |
+-----+-----+-----+-----+
```

We do NOT choose one random square and declare it the prior. We score all candidate squares. The goal is to create a probability landscape: cells that are more plausible get larger prior mass; implausible cells get less.

Why a grid?

- It naturally represents uncertainty over space instead of forcing everything into one point.
- It can handle irregular or even multi-peaked posterior beliefs.
- It works well with spatial masks such as town/locality/landmark constraints.
- It lets us obtain a confidence region by accumulating posterior probability over cells.

Candidate location variable

$$L = \text{one candidate location / grid cell}$$

The Bayesian equation will ultimately assign a posterior score to every candidate L. The highest-posterior location becomes the predicted pin, while the full set of cell probabilities represents the uncertainty around it.

Practical caution

The grid resolution is a design choice. Finer grids give more spatial detail but cost more computation; coarse grids may blur small location differences. The existing report uses 25 m as an example.

4. Step 2 - build the prior from the geocoder and spatial clues

The geocoder gives a point g = (g_x, g_y) and a precision label. We use it as a starting spatial belief, not as ground truth.

$$P(L) \propto \exp(- ||L - g||^2 / (2 \sigma_prec^2))$$

This comes directly from a 2D isotropic Gaussian density centred at the geocoder point. We ignore the normalizing constant when comparing relative prior scores across grid cells.

Meaning of σ_prec

σ_prec is the learned uncertainty/width associated with a geocoder precision category. It answers: "How widely should I spread probability around the geocoder point for this precision level?"

Precision	Qualitative treatment	Data caveat
rooftop	Very narrow prior in principle	Only 1 surveyed example - do not learn a reliable rooftop-specific sigma from n=1.
street	Narrower than locality/pincode	Enough data to estimate a useful empirical width.
locality	Broad prior around geocoder	Most common category; its error is roughly hundreds of metres.
pincode	Very broad prior	Error is kilometre-scale; prior must be wide.

Other evidence that can reshape the prior

- Town extent: restrict probability from spilling outside a plausible town boundary.
- Locality centroid: use a known locality centre when the upstream parser identifies a locality.
- Landmark locations: pull or constrain the prior around a referenced landmark, if the upstream parser provides one.
- Nearby confirmed addresses: use trusted neighbouring locations as spatial context, with careful leakage control.
- Address type: office/residence/permanent_native may behave differently and can be treated as a category in the prior model.

5. Step 3 - a field visit becomes a likelihood

For a visit i, let z_i = (checkin_x, checkin_y) be the observed GPS point. For each candidate location L, ask: “If the true house were at L, how plausible is it that the agent would check in at z_i?”

$$\text{Likelihood} = P(V_i \mid L)$$

For a genuine visit, the basic model is Gaussian around L. The farther the check-in is from L, the smaller the likelihood.

$$\text{Gaussian score} \propto \exp(- d^2 / (2\ \sigma_i^2))$$

where d = distance(z_i, L). The report illustrates the idea with a genuine-visit sigma of 20 m: at 10 m the Gaussian score is about 0.88, at 40 m about 0.14, and at 100 m it is almost zero.

What the likelihood is NOT

- It is not a probability directly written in the CSV.
- It is not “probability that the visit happened”.
- It is not a single number independent of L.
- It is a compatibility/density score: how well the observed visit agrees with a particular candidate location.

Why this matters

One GPS point can support some cells strongly and other cells weakly. That is how the visit actually changes the spatial probability map.

6. Step 4 - add visit trust and the outlier mixture

The dataset contains visits that are often excellent but sometimes wildly wrong. So we do not force every visit to behave like a genuine Gaussian measurement.

The trust variable

$$\pi_i = P(\text{visit } i \text{ is genuine})$$

π_i is a model-derived reliability score. It is NOT the recorded outcome. For example, “met_borrower” is an observed outcome in the data; $\pi_i = 0.93$ would be our estimated trust.

The mixture likelihood

$$p(z_i \mid L) = \pi_i \, N(z_i; L, \sigma_i^2 I) + (1 - \pi_i) \, \text{Outlier}(z_i)$$

Symbol	Meaning
z_i	Observed GPS check-in for visit i.
L	Candidate house location / grid cell.
π_i	How much we trust this visit as a genuine location observation.
$1 - \pi_i$	Remaining mass assigned to the possibility that the observation is an outlier.
σ_i	Uncertainty/spread of the genuine-visit GPS error.
$N(...)$	Gaussian likelihood for a genuine visit.
$\text{Outlier}(z_i)$	Model for a suspicious/bad observation.

Why mix the two?

Suppose four visits cluster near the house and one is 2 km away. A pure Gaussian treats the far point as an impossible-looking measurement. The mixture says: “It might simply be junk; do not let it destroy the inference.”

What can influence π_i?

Possible signals from the available visit/agent data include outcome, GPS accuracy, repeated photo hashes, agent hotspot behaviour, GPS trail displacement/path, remark contradictions, agreement with other visits, and agent context. The supplied report explicitly cautions that there are no proven fraud labels, so the output is a reliability score, not a certified fraud detector.

7. Step 5 - estimate σ_i for each visit

σ_i answers a practical question: “If this agent/device stood at the same door repeatedly, how much would the recorded check-in scatter?”

Why GPS accuracy alone is not enough

The visit data shows that reported GPS accuracy is useful, but actual location error is typically larger. The median ratio of error to gps_accuracy_m is about 2.4 in the supplied data.

$$\sigma_i = \mathrm{sqrt}(a^2 + (b \times \mathrm{accuracy_i})^2)$$

The report proposes learning the parameters a and b from the training addresses. a is a floor for irreducible door-level scatter; b maps reported GPS accuracy to observed location-error scale.

Two different sigmas - do not mix them

Parameter	Used for	Meaning
σ_prec	Geocoder prior	Uncertainty associated with a geocoder precision category.
σ_i	Individual visit likelihood	Uncertainty of this particular GPS visit.
σ_post / final spread	Combined belief	Uncertainty after combining prior + visit evidence; depends on the chosen inference representation.

What data informs σ_i?

- accuracy_m / gps_accuracy_m from visit/GPS records
- actual errors on the 100 surveyed addresses for validating/calibrating the mapping
- potential conditioning by outcome or other visit-quality signals if there is enough data
- careful separation between training, validation and test so the true errors of test addresses do not leak into sigma estimation.

Rooftop reminder

The one surveyed rooftop case is not enough to estimate a reliable rooftop-specific prior width. Treat the rooftop precision label as a signal, but use smoothing/borrowing/conservative fallback rather than pretending n=1 is a stable distribution.

8. Step 6 - Bayesian fusion: combine all the evidence

$$P(L \mid E) \propto P(L) \times \prod_{i=1}^{N} P(V_i \mid L)$$

For every candidate cell L, we multiply its prior score by the likelihood supplied by each visit. Then we normalize the scores so that the posterior represents a probability distribution over candidate locations.

What each term means in our project

Term	Project meaning
P(L)	Prior belief for this grid cell from geocoder and spatial clues.
P(V_i | L)	Compatibility of visit i with this candidate cell.
Product over i	Combine all visit evidence under the conditional-independence approximation.
P(L | E)	Posterior belief: after seeing all evidence, how plausible is this candidate location?

Mini numerical example

Candidate	Prior	Visit 1 likelihood	Visit 2 likelihood	Unnormalised score
A	0.50	0.90	0.80	0.360
B	0.30	0.30	0.20	0.018
C	0.20	0.10	0.40	0.008

After normalization, A gets almost all the posterior mass because both visits agree with it. This is the core Bayesian idea: the prior gives a starting belief; consistent evidence multiplies that belief upward, while conflicting evidence pushes it down.

The assumption behind the product

The product form assumes that, once the true location is fixed, the individual visits are approximately conditionally independent. This is a modelling approximation. Repeated visits by the same agent or correlated behaviour can violate it, which is one reason trust weighting and robust likelihoods matter.

9. Step 7 - Bayesian, Gaussian, sigma and radius are four different jobs

Concept	Job in our project
Bayesian inference	Combines different evidence sources and produces the posterior over candidate locations.
Gaussian distribution	One mathematical model for how genuine GPS errors are spread around a candidate location.
Sigma (σ)	Controls how wide that Gaussian uncertainty is.
Confidence radius	Summarizes how much spatial area we need around the predicted point to cover a chosen level of probability/coverage.

Bayesian fusion → posterior location distribution
Gaussian model → one possible shape for uncertainty
Sigma → width/spread of that shape
Radius → selected spatial region around the pin

Most important distinction
Bayes does NOT require Gaussian. Gaussian is a convenient likelihood/uncertainty assumption. A grid posterior can represent probability over space without forcing the final belief into one Gaussian bump.

What the posterior gives us

The posterior is a probability landscape over the map, not just one coordinate. The predicted location can be the highest-posterior cell (the pin). The spread or geometry of the posterior tells us how uncertain the prediction is.

Why the final system prefers a grid

A grid can represent irregular, skewed or multi-cluster beliefs. If two different neighbourhoods remain plausible, a grid can keep both modes alive instead of forcing everything into one bell-shaped cloud.

10. Step 8 - why a pure Gaussian is not enough

The observed error distribution is not nicely bell-shaped. Most successful visits are close, but a small number can be kilometres away. This creates heavy tails and outliers.

Situation	What the data suggests	Implication
Good door visit	Typical error tens of metres	Gaussian-like core can be useful.
Rare large error	Some visits are 1-8 km away	Pure Gaussian can become overconfident / under-cover.
Suspicious visit	Repeated photos / hotspot patterns exist	Need a trust or outlier mechanism.
Conflicting visits	Different location clusters can occur	A single Gaussian bump may be the wrong shape.
No visits	Only prior/spatial clues exist	Need a different evidence construction than a visit likelihood.

Better likelihood options to investigate

Option	Why consider it?
Gaussian + outlier mixture	Keeps a clean Gaussian core but protects against junk visits.
Student-t	Heavier tails than Gaussian; naturally more tolerant of moderate outliers.
Empirical error distribution	Uses observed training errors directly; fewer distributional assumptions but needs enough training error data.
KDE	Builds a smooth density from observations; useful if evidence is naturally multimodal.
Grid posterior	Places probability directly on spatial cells; handles irregular/multimodal shapes well.
Robust likelihood (Huber/M-type)	Reduces the impact of large residuals without declaring a hard outlier.

Research direction

The strongest refinement is not “replace Gaussian with something fancy” for its own sake. It is: keep a spatial posterior, then use the simplest likelihood that matches the observed error tails and visit reliability.

11. Step 9 - alternative ways to predict the location itself

A. Robust spatial estimators

Method	Core idea	Main weakness
Mean	Average all check-ins.	One bad visit can drag the point far away.
Coordinate median	Take median x and median y.	Can create a “between” point that is not geometrically representative.
Geometric median	Point minimizing total distance to visits.	Still depends on majority of visits being useful.
Weighted geometric median	Same, but trusted/accurate visits get more weight.	Needs a good reliability/weight estimate.
RANSAC	Find a consistent subset and fit to it.	Needs enough visits; sparse cases are hard.
M-estimator	Down-weight large residuals smoothly.	Requires a threshold/robustness choice.

B. Clustering

DBSCAN/HDBSCAN can identify dense groups of GPS points and treat isolated points as noise. This is useful when four visits cluster around one location and one visit sits kilometres away.

Visits 1-4: ● ● ● ● → dense cluster
Visit 5: × → noise / possible outlier

Caution: many visited addresses have only 1-3 visits. Clustering becomes less reliable when there are too few points to form a stable cluster.

When to prefer which?

- Few visits + mostly clean data: weighted/geometric median is a strong simple baseline.
- Several visits + clear dense cluster: HDBSCAN/DBSCAN is informative.
- Messy visits + evidence from the geocoder: Bayesian/grid fusion can combine both instead of discarding either source.
- Very small supervised sample: avoid deep models that require large labelled datasets.

12. Step 10 - residual learning and pseudo-labels

A promising alternative is to treat the commercial geocoder as a baseline that already knows something about the location, and ask ML to learn the correction instead of predicting the whole coordinate from scratch.

Address-derived features + geocoder
↓
baseline point g
↓
predict correction Δx, Δy
↓
final location = g + Δ

Why this could be easier

- The geocoder already provides a reasonable starting point, especially at street/locality scale.
- The learning problem becomes “how far and in what direction is the geocoder usually wrong here?”
- Useful features can include precision, town_id, address_type, structured locality/landmark features, landmark distances/directions, and local visit-based corrections.
- This is especially relevant for the no-visit population, where the geocoder is the main point of reference.

The bigger data opportunity - pseudo-labels

Only 66 verified locations are available for supervised fitting, but the dataset contains many more addresses and 1,477 addresses with at least one visit. For an unlabelled address with several trustworthy, mutually consistent visits, we can create an approximate location consensus and use it as a pseudo-label for exploratory learning.

Critical warning
A pseudo-label is not ground truth. If the visits are biased or suspicious, the pseudo-label inherits that bias. The trust filter must come before pseudo-label generation, and pseudo-label experiments must be evaluated against the 100 verified addresses.

Potential hybrid

Geocoder residual correction can improve the prior centre; robust visit evidence can then update that centre; the spatial posterior can combine both. The goal is not to make every method independent - it is to let each method fix a weakness of another.

13. Step 11 - no-visit branch: Convex Hull + fuzzy boundary

This is the refinement for the cold-start case where len(visits_data) = 0. Instead of using only the geocoder point, collect all available spatial clues supplied by the upstream parser and the spatial tables.

Evidence points

Evidence point	Source	Interpretation
Geocoder point	baseline_geocodes.csv	Baseline centre; uncertainty depends on precision.
Locality centroid	localities.csv	A clue for where the address lies within the known locality.
Landmark / POI	landmarks_poi.csv	A concrete spatial reference; useful when the parsed address identifies it.

Construct a set of points P1, P2, ... , Pn. Then compute their convex hull: the smallest polygon containing all evidence points.

Geocoder point ● Locality centroid ●
● POI
↘️ stretch a boundary around the outermost points ↙️
CONVEX HULL

Why not just average all evidence points?

A mean can be dominated by clue count. If five landmarks lie near one spot and one locality centroid is elsewhere, the mean can be pulled toward the landmarks. The hull treats the evidence geometry more neutrally and preserves the spread of the available clues.

But the hull alone is not a confidence interval

The hull is a feasible/geometry-based region. We still need a “fuzzy” boundary that says how much probability can bleed outside each edge based on the uncertainty of the two points forming that edge.

14. Step 12 - fuzzy hull confidence radius for the no-visit case

For each hull edge, take the two endpoints A and B and their uncertainties σ_A and σ_B. The report proposes an uncertainty for the edge based on those endpoint uncertainties, roughly an inverse-variance weighted combination, with the close-uncertainty case behaving approximately like an average.

$$\sigma_edge \approx (\sigma_A + \sigma_B) / 2 \quad \text{[simple close-uncertainty intuition]}$$

Turn the edge into a fuzzy boundary

Imagine standing on an edge and looking outward. Instead of saying the boundary is perfectly sharp, model a normal distribution perpendicular to that edge: most belief stays near the edge, but some mass can extend outward.

$$D_edge = z_\alpha \times \sigma_edge$$

Here z_α is a one-sided normal quantile. The report uses approximately: 80% → 0.842, 90% → 1.282, 95% → 1.645.

Expand each hull edge

Push each edge outward, away from the hull centroid, by D_edge. The collection of these expanded edges forms a "Confidence Hull".

Original hull edge ————————
↑ D_edge
Expanded edge ———————— (pushed outward)

Final raw radius

$$R_\alpha = \text{max distance from hull centroid to a vertex of the Confidence Hull}$$

The hull centroid is the predicted point for this branch. The radius is conservative: it is the maximum distance needed to cover the confidence hull from that predicted point.

Fallbacks

If there is only 1 evidence point, use that point as the prediction and its sigma directly as the radius. If there are 2 points, use the midpoint and a widened radius around the segment. With 3+ non-collinear points, run the full hull procedure.

15. Step 13 - conformal calibration: make the confidence claim honest

A raw Gaussian radius or fuzzy-hull radius comes from modelling assumptions. Conformal calibration is a separate data-driven layer that checks whether the claimed coverage actually holds on held-out addresses with known truth.

The definition of the desired region

$$P(||\text{true} - \text{pin}|| \leq r_\alpha) = \alpha$$

For α = 0.80, the goal is an 80% prediction region that contains the true location in about 80% of comparable cases.

Split-conformal idea

- Fit the location/radius model using the training addresses.
- Use a separate calibration set (the report suggests the 19 validation rows) to measure how the raw radius behaves.
- For each calibration address, compute the actual error from predicted pin to true location and compare that error with the raw radius.
- Use a suitable quantile of those scores as a calibration multiplier q.
- For a new address: final_radius = q × raw_radius.

$$\text{score}_i = \text{error}_i / \text{raw_radius}_i$$

$$\text{final radius} = q_\alpha \times \text{raw radius}$$

Why this is valuable

If the Gaussian model is too optimistic, q > 1 stretches the region. If it is already well calibrated, q can be near 1. The key is that the multiplier is learned from held-out truth rather than invented.

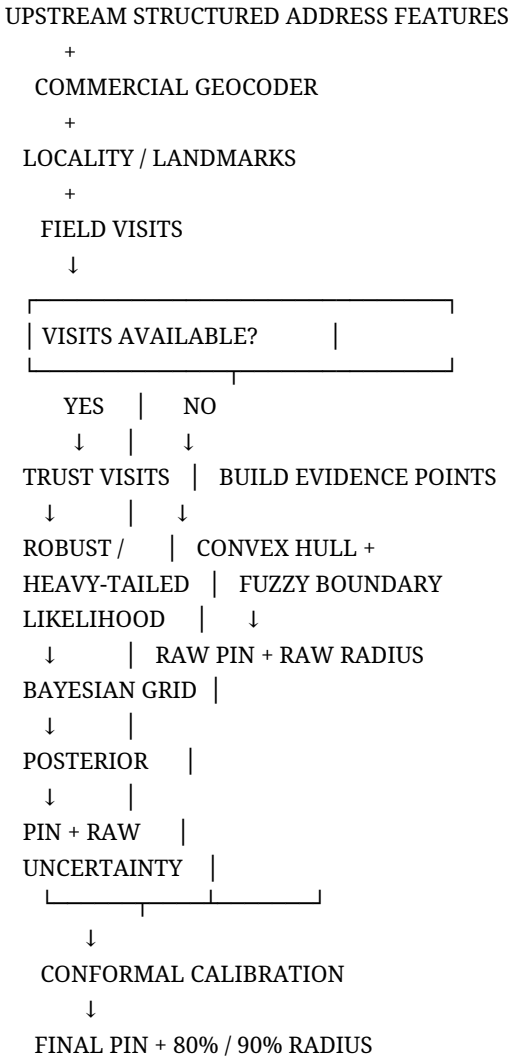
Group-aware calibration

A single multiplier for everybody may be poor when visited and no-visit cases behave very differently. Mondrian/group-aware conformal can use separate calibration groups such as “visits available vs no visits” or carefully justified geocoder/reliability regimes, provided each group has enough calibration data.

Small-data warning

There are only 19 validation addresses. Quantiles from 19 observations are noisy. Report calibration uncertainty honestly; do not over-interpret one exact 80% number.

16. Final consolidated workflow and what we should actually test



What I would treat as the core methodology

- Visited branch: Bayesian grid fusion with a geocoder-based prior, visit-specific uncertainty, visit trust, and an outlier/heavy-tailed likelihood.
- No-visit branch: geocoder + locality + landmark evidence assembled into a spatial region; use convex-hull/fuzzy-boundary reasoning as a practical geometry-based prior/uncertainty method.
- Final uncertainty: conformal calibration on held-out verified addresses, with group-aware calibration considered when sample size permits.

What should be compared before calling anything "best"

Experiment	Why
Commercial geocoder baseline	Defines the score to beat.
Median / geometric median / weighted geometric median	Simple robust baselines for visited addresses.
DBSCAN/HDBSCAN	Tests whether cluster structure is useful for outliers/multiple visit modes.
Bayesian + Gaussian/outlier	Current principled evidence-fusion candidate.
Bayesian + robust/heavy-tailed likelihood	Tests whether the observed heavy tails are handled better.
Geocoder residual ML	Tests whether learning correction Δx, Δy is easier than learning absolute location.
No-visit convex hull branch vs prior-only	Tests whether locality/landmark geometry actually improves cold-start predictions.
Conformal calibration	Tests whether the final 80% / 90% regions achieve their promised coverage.

Final takeaway

The project should not be “a Gaussian wrapped around a geocoder”. The robust story is: use the geocoder as an uncertainty-aware starting point, let trustworthy visits move the belief toward where agents actually reached, protect the inference from bad evidence, use spatial constraints when visits are absent, and calibrate the final region against real held-out locations.

Most important implementation principle

Every parameter that affects confidence - prior width, visit sigma, trust, and calibration - should be learned or checked without leaking the test set. The verified 100-address set is small, so simple, robust and well-calibrated beats unnecessarily complicated.